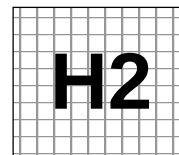


Civics Group	Index Number	Name (use BLOCK LETTERS)
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ST ANDREW'S JUNIOR COLLEGE
2016 JC2 Preliminary Examinations

H2 BIOLOGY

9648/3

Paper 3: Applications and Planning Question

Tuesday

13th September 2016

2 hours

Additional Materials: Answer Paper
 Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your civics group, index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination,

1. Attach Question 5 to the **cover sheet** provided.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Paper 3	
1	/12
2	/17
3	/11
Total	/40
4 (Planning)	/12

This document consists of **21** printed pages.

[Turn over

Answer **all** questions.

QUESTION 1

Tissue plasminogen activator (TPA) is a blood protein that scavenges and dissolves blood clots that may form in the final stages of the healing process. TPA is primarily used in heart patients or others suffering from poor circulation to prevent the development of clots that can be life-threatening. Hence, microbially produced TPA is in high demand.

Fig. 1.1 shows a commercially available plasmid that can be used to clone a synthetic TPA gene into *E. coli*. It contains two selectable markers which confer resistance to the antibiotics vancomycin (Van^R gene) and ampicillin (Amp^R gene) respectively.

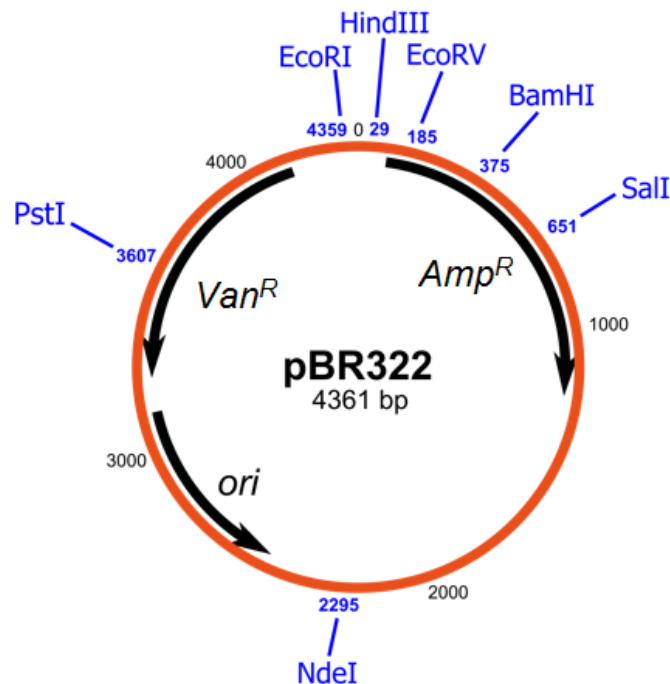


Fig. 1.1

(a)(i) Describe the natural function of restriction enzymes.

.....[1]

- 1 They protect bacteria from attack by viruses/bacteriophages by cleaving any foreign DNA that enters bacterial cell

(ii) *Bam*H1 was used to produce recombinant plasmids containing TPA gene. Describe and explain the selection procedure that would allow for selection of bacteria which contain the recombinant plasmids.

-[5]
- 1 [Procedure] Plate the bacteria onto **LB/van plate** (and incubate it overnight)
 - 2 [Explain] **Transformed cells carrying the plasmid** would survive in vancomycin medium as the plasmid contains the intact *Van^R* gene, **functional protein produced** which confers the cells with resistance to vancomycin
 - 3 [Procedure] A replica plate is then made using a piece of nitrocellulose filter paper to transfer bacterial colonies from the LB/van plate onto a LB/amp plate, (which is incubated overnight).
 - 4 [Procedure] The LB/van and LB/amp plates are then compared, and the colonies which grew on LB/van plates but not on LB/amp plates are those which contain recombinant plasmids.
 - 5 [Explain] This is because the insertion of TPA gene into amp gene disrupts the Amp^R gene, **no functional protein produced** to confer resistance to ampicillin.

(iii) The TPA gene which is inserted into the plasmid can be artificially synthesised. This synthetic TPA gene inserted into the plasmid may possess a different nucleotide sequence to the actual human gene. Explain one reason why this is so.

-[2]
- 1 The sequence of the synthetic gene which is **derived from the amino acid sequence** may be different from the actual human gene.
 - 2 ref. genetic code being degenerate / some amino acids are encoded by more than one codon;

OR

- 3 The sequence of the synthetic gene which is **derived from the amino acid sequence** may be different from the actual human gene
- 4 The synthetic TPA gene **does not contain introns**

(b) Discuss one problem of producing eukaryotic proteins using a prokaryotic host and how it can be overcome.

.....[3]

[Problem]

- 1 Prokaryotic hosts do not have spliceosomes and cannot remove introns in eukaryotic genes
- 2 Introns will be translated, causing the wrong amino acids to be incorporated into the protein or presence of stop codon in intron leads to premature termination of translation.

[Solution]

- 3 Synthesize the cDNA of the gene from the mRNA using reverse transcriptase/
/ Use of eukaryotic cells such as yeast or mammalian cells which can perform RNA splicing

OR

[Problem]

- 4 Prokaryotic hosts do not have Golgi body and RER and cannot perform post-translational modifications (such as glycosylation)
- 5 The polypeptides produced may not be biologically active.

[Solution]

- 6 Use of eukaryotic cells such as yeast or mammalian cells as hosts for cloning and/or expressing eukaryotic genes of interest.

OR

[Problem]

- 7 Only eukaryotic promoters (and other DNA control sequences) present / no prokaryotic promoters present
- 8 Bacterial cells may not be able to recognize eukaryotic promoter sequences (and other control elements)

[Solution]

- 9 Use an expression vector which contains a prokaryotic promoter (and any other control elements) necessary for the gene's transcription (and translation).

(c) Two DNA samples of insulin gene were obtained using different methods:

1st method: reverse transcription of insulin mRNA to produce cDNA

2nd method: extraction of genomic DNA followed by selective amplification of the insulin gene using PCR

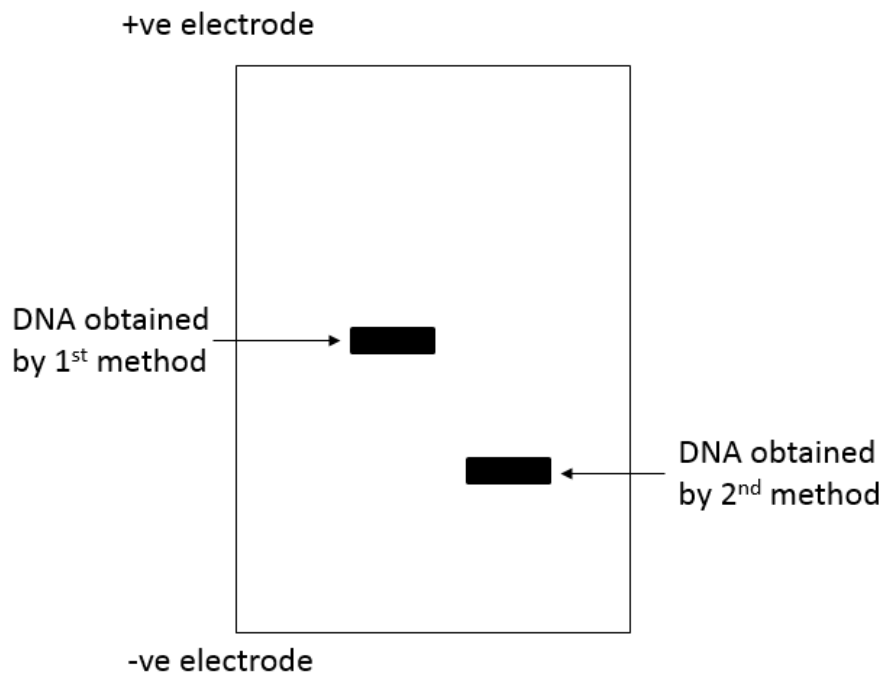
The two DNA samples were loaded into **different** wells and separated via **gel electrophoresis**. Draw the band(s) you would expect to see for the DNA sample obtained using the 2nd method in Fig 1.2. [1]

5



Fig. 1.2

ANS:



[Q1 Total: 12]

QUESTION 2

Individuals born with severe combined immune deficiency (SCID) have no defence against common infections. This disease is caused by the possession of recessive alleles.

Fig.2.1 shows the pedigree which traces the inheritance of SCID in a family.

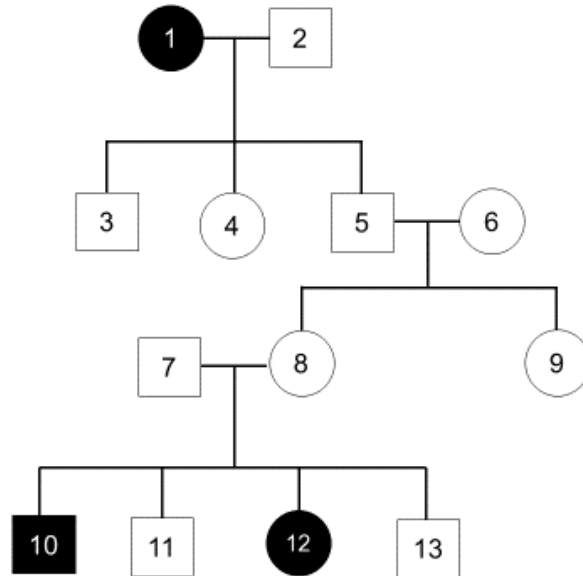


Fig. 2.1

(a)(i) Explain if the SCID tracked in Fig. 2.1 is autosomal or X-linked.

-[1]
- 1 Autosomal; diseased mother #1 did not give rise to diseased sons #3 and 5;
/ female offspring #12 is diseased while father #7 is not diseased (thus, not X-linked)

(ii) Explain the genetic basis behind the susceptibility to common infections in this mode of SCID.

-[3]
- 1 Autosomal recessive mutation in adenosine deaminase (ADA) gene on chromosome 20;
 - 2 Deficiency of **functional** enzyme ADA which catalyses the deamination of adenosine and deoxyadenosine;
 - 3 Adenosine and deoxyadenosine **accumulation** is toxic to lymphocytes;;
 - 4 Replication of lymphocytes of lymphocytes is blocked, leaving insufficient T and B lymphocytes to fight off infections;
Immune system is compromised;
(Therefore, patient is susceptible to opportunistic infections)

(iii) Explain why SCID is a suitable candidate for treatment using gene therapy.

.....[2]

- 1 SCID is a recessive disease (in one gene);
caused by possession of 2 recessive alleles (in one gene);
- 2 Inserted normal allele is dominant;
one copy is enough to produce the functional gene product/protein (to mask effects of recessive mutation);

Gene therapy is carried out for individual 1 (shown in Fig.2.1). The essential steps are summarized in Fig.2.2.

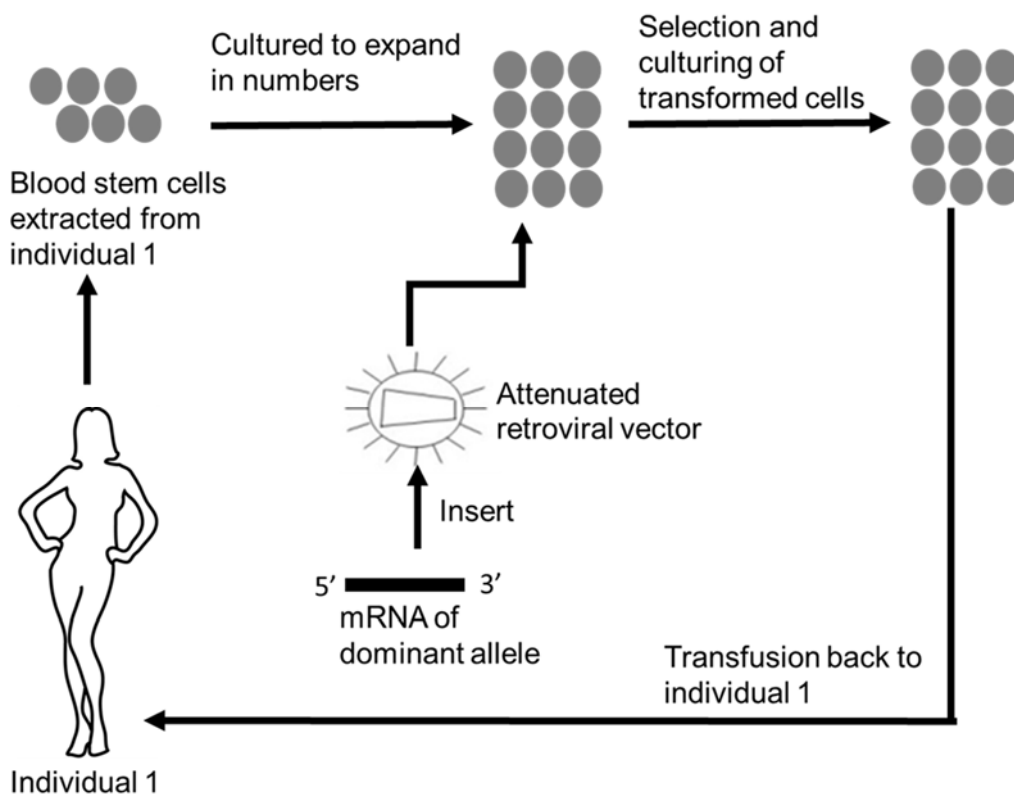


Fig.2.2

(iv) Give two reasons why blood stem cells are chosen to receive the dominant allele.

.....[2]

- 1 multipotent;
able to differentiate into specialized blood cell types / give rise to a limited range of cell type / produce only cells of a specific lineage
e.g. T and B lymphocytes to replace disease cells lacking ADA in SCID;
- 2 capable of self-renewal / dividing rapidly by mitosis;
ref. asymmetric division to generate continuous supply of terminally differentiated cells for long-term treatment;

(v) Suggest why the *ex vivo* method was used in this case.

.....[1]

- 1 blood stem cells can be easily extracted from **bone marrow** and **isolated**
- 2 ref. easily **cultured** in the laboratory to **increase in numbers**;
- 3 allow for **selection** of successfully transformed cells ;

[Any 1]

Answers related to the usage of viral vectors are rejected as the focus is on the usage of *ex vivo* method.

(vi) The gene therapy treatment described in Fig.2.2 turned out to be ineffective in treating individual. Suggest two reasons why.

.....[2]

- 1 Immune response against the cells containing viral glycoproteins in the plasma membranes (not the virus) in the body;;
- 2 Difficulties in controlling the activity of the transferred gene: cells must make appropriate amounts of the gene product at the right time and in the right place

(vii) Explain why individuals treated **successfully** with gene therapy do not pass on the dominant allele to their children.

.....[1]

- 1 Dominant allele inserted in blood stem cells which are somatic cells;;
/ Only genetic alteration of reproductive / germ cells will result in normal allele be passed to the offspring;;

(viii) Discuss two ethical considerations for the use of gene therapy.

.....[2]

Safety (Max 1)

- 1 Question of **safety and efficiency** of treatment and therefore **unknown long term consequences** to the well-being of the recipients.
- 2 Question of **safety and efficiency** of treatment , raises questions over who should participate in a human trial – the terminally ill with no treatment options or anyone who understands the potential risk but is a willing patient.

Privacy

- 3 Concerns about the protection of privacy and confidentiality of medical information of patients involved in clinical trials which may have implications on (medical) insurance coverage / employability etc.

Interfering with life

- 4 The unborn child who would be affected by germline gene therapy cannot choose whether to have the treatment.

[Any 2 ideas]

AVP: (max 1 per idea)

Interfere with evolution

- 1 Elimination of unwanted alleles from the gene pool will decrease genetic variation.
- 2 Detrimental to the survival of our species in future as genes that are damaging under some conditions may be advantageous under other conditions.

- 3 Germ-line gene therapy would forever change the genetic makeup of an individual's descendants / the human gene pool would be permanently affected.

Cost

- 4 Gene therapy is currently very expensive, require expertise and equipment found only in major medical centres, concerns about gene therapy become a luxury available only to the rich and powerful.
- 5 Possible genetic enhancements creating an advantage for those who can afford the treatment.

Eugenics social policies

- 6 Type of circumstances that determines whether genomes should be altered/ what is normal or a disability or disorder, and who decides.
- 7 Potential for non-therapeutic enhancement possibilities.
- 8 Concerns about the widespread use of gene therapy making society less accepting of people who have mild disorders / genetic diseases or less able / lead to new definitions of "normal (e.g. intellect / height / strength etc.)

It is possible to diagnose for SCID in the individuals shown in Fig. 2.1, by detecting known RFLP regions found **within** the gene.

(ix) Explain how the Human Genome Project facilitates this process.

-[1]
 1 **Probes complementary to the RFLP regions** can be designed with **knowledge of the DNA sequences of the human genome**;;

In SCID, a mutation alters one base of the coding sequence, resulting in a loss of a restriction site. The gene responsible for SCID is isolated from **individual 8** of Fig. 2.1. Restriction digestion with *MspI* enzyme results in 0.47 kb, 1.834 kb and 2.304 kb fragments.

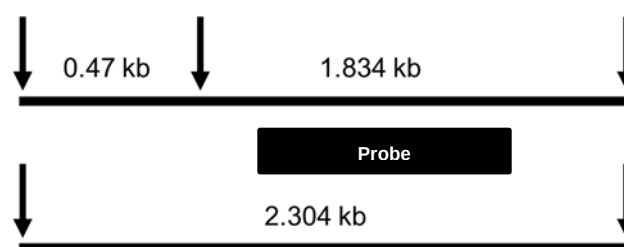
(x) The lines in Fig. 2.3 represent the RFLP locus on the homologous chromosomes. Draw the restriction map showing the **cut sites** of *MspI* enzyme, **using arrows**, for individual 8. The **size** of the fragments between cut sites should be indicated.

.....[1]



Fig. 2.3

Ans:



A DNA probe is added which resulted in the detection of only the 1.834 kb and 2.304 kb bands.

(xi) Draw the suggested position of the probe in Fig. 2.3.

.....[1]

- 1 Anywhere within the 1.834 kb fragment region

[Q2 Total: 17]

QUESTION 3

In Singapore, orchids of superior quality are selected for mass production through tissue culture.

Tissue culture is a collection of techniques used to grow plant cells, tissues, or organs under sterile conditions on a nutrient culture medium.

- (a) Equal masses of orchid plant callus were cultured on media containing different concentrations of two plant growth regulators: auxin and cytokinin. The results are shown in Table 3.1.

treatment	concentration of plant growth regulators / mg dm^{-3}		effect of plant growth regulators on the callus
	auxin	cytokinin	
A	2.00	0.00	little or no growth
B	2.00	0.02	growth of roots
C	2.00	0.20	increased growth of callus with no differentiation
D	2.00	0.50	growth of shoots
E	0.00	0.20	little or no growth

Table 3.1

With reference to Table 3.1,

- (i) describe the effects of auxin and cytokinin on the callus;

.....[3]

- Both cytokinin and auxin are needed for cell division/growth / little or no growth if only either cytokinin or auxin is present; as seen in treatment A with (2.00 mg dm^{-3} auxin and) 0.00 mg dm^{-3} cytokinin + treatment E with 0.00 mg dm^{-3} auxin (and 0.20 mg dm^{-3} cytokinin);
- (Treatment B) **High** auxin : cytokinin **ratio** of $2.00 : 0.02$ / Lowest concentration of cytokinin at 0.02 mg dm^{-3} compared to auxin at 2.00 mg dm^{-3} results only in root growth;
- (Treatment D) **Low** auxin : cytokinin **ratio** of $2.00 : 0.50$ / Highest concentration of cytokinin at 0.50 mg dm^{-3} compared to auxin at 2.00 mg dm^{-3} , results only in shoot growth;
- (Treatment C) **Intermediate** auxin : cytokinin **ratio** of $2.00 : 0.20$ / Intermediate cytokinin to auxin concentration, results in increased growth of callus with no differentiation;
- Auxin is required for growth at all cases (with reference to treatment E);

(ii) Suggest what treatment should be given to the shoots developed from callus in treatment D to turn them into plantlets.

.....[1]
 1 **Transfer** the shoots from treatment D to a medium containing 0.02 mg dm^{-3} cytokinin and 2.00 mg dm^{-3} auxin to stimulate **root growth**.

(b) Explain why plantlets grown from a callus in tissue culture are considered to be clones of each other.

.....[2]
 1 A callus is a mass of (undifferentiated cells) as a result of **mitosis** of the cells of the plant tissue it is derived from;
 / The plantlets are grown as a result of asexual reproduction;
 2 resulting in the plantlets being genetically identical/having the same genotype as the plant from which the callus is derived;

(c) Tissue culture is also used to grow genetically engineered plant cells into whole plants. An example are crops which are genetically engineered to carry the Bt toxin gene from the bacteria *Bacillus thuringiensis*.

Explain the benefit of incorporating a Bt gene into a crop plant.

.....[2]
 1 Bt toxin is **synthesized**, kill the pest that consumes it/plant is pest resistant;
 2 Reduce crop losses / Increase crop yield;

(d) The genetically modified (GM) salmon was created in 1992 by Emeritus Professor Hew Choy Leong from the department of biological sciences at the National University of Singapore.

Explain why this development of GM salmon is supported by some.

.....[3]
 1 Atlantic salmon with more active salmon growth hormone gene has **active growth hormone gene from the Pacific chinook salmon** and **regulatory elements of the ocean pout**
 2 Grow to its desired market length/full length in a shorter period of time;
 3 Increasing yield and greater profits per unit time;
 4 Meeting the world's food needs with a declining fish population;

AVP: (Max 1)

- 1 Accidental release of transgenic organisms into the environment might upset the balance of the ecosystem.
- 2 Fast-growing salmon may outcompete the wild salmon population and affect the food chain.
- 3 Larger transgenic salmon may be preferably selected as mates over smaller wild types.
- 4 Danger that the active growth hormone gene is transferred to other fish.

[Q3 Total: 11]

QUESTION 4**Planning question**

You are required to plan, but not carry out, an investigation into the effect of different temperatures on the rate of respiration in crickets.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you must use:

- Crickets
- Manometer with scale (internal radius of manometer is 1mm)
- Manometer fluid
- 2 rubber bungs with delivery tubes
- 2 boiling tubes
- Zinc gauze platform (to provide platform to support the crickets)
- Glass beads
- Clamps and stands
- Stop watch
- Soda lime solution
- Thermometer
- Supply of water at 50°C
- Tap water / ice
- Weighing balance
- A variety of different sized beakers, measuring cylinders, and syringes for measuring volumes

Your plan should have a clear and helpful structure to include:

- an explanation of theory to support your practical procedure,
- a description of the method used, including the scientific reasoning behind the method and any recommended safety measures,
- an explanation of dependent and independent variables involved,
- relevant and clearly labeled diagrams,
- how you will record your results and ensure they are accurate and reliable as possible,
- proposed layout of results tables and graphs with clear headings and labels,
- the correct use of technical and scientific terms.

[Q4 Total: 12]

MARK SCHEME**Suggested write-up****1 Theoretical consideration or rationale of the plan to justify the practical procedure (2 marks)****(3 and below: 0 marks; 4-5 pts - 1 mark, 6-8 pts – 2 marks)****[Background]**

- a) Ref to enzymes involved in reactions that occur in glycolysis/the link pathway/Krebs cycle/oxidative phosphorylation.
- b) O₂ is used as the final electron acceptor in the ETC. The protons and electrons recombine with oxygen to form water.
- c) CO₂ is produced during oxidative decarboxylation during the link reaction and the Krebs cycle

[Hypothesis]

- d) Higher rate of O₂ uptake indicates a higher rate of respiration. At optimum temperature, rate of respiration is highest.
- e) As temperature increases, kinetic energy increases, more ES complexes formed, rate of respiration increases.
- f) When temperatures are too high, 3D structure of enzymes involved in respiration disrupted, enzymes are denatured, rate of respiration decreases.

[Rationale]

- g) Soda lime absorbs any CO₂ produced by the crickets. The pressure in the vessel decreases due to the uptake of O₂ during respiration and air is sucked from the capillary to keep the pressure constant.
- h) The oxygen uptake is detected by the displacement of manometer fluid.

[1 mark for independent and dependent variable]**Independent variable**

Temperatures of water bath maintained at 30°C, 35°C, 40°C, 45°C, 50°C (max at 50°C)

+

Dependent variable

Respiration rate = **rate** of oxygen taken up (mm³ / min) which is calculated by dividing volume of oxygen by time

- Volume of oxygen = $\pi r^2 \times d$
- r = internal radius of manometer tube
- d = displacement by manometer fluid

[1 mark]**Variables to be kept constant**

- Mass of cricket, using weighing balance
- Volume of soda lime, using weighing balance
- Duration of measurements, using stop watch

[1 mark Control]

Place cricket(s) of similar mass 5g in the experimental tube in the control tube. The cricket must not come into contact with the soda-lime. Rationale: to ensure that any displacement of manometer fluid is due to respiration of the crickets.

Sample Procedure

1. Half fill a manometer with fluid and connect both arms of the manometer to rubber bungs with delivery tubes.

2. Add 5 cm³ of soda lime using syringe / measuring cylinder into two boiling tubes and then place a zinc-gauge platform 1 cm above the soda-lime. One of the boiling tube serves as an experimental tube (respiration chamber) and the other the compensation tube.

3. Prepare a large beaker of water, into which both the experimental tube and control tube can be submerged. Maintain the temperature of the water at a constant temperature of 30°C using hot and cold water. Monitor the temperature with a thermometer.

4. Prepare 5g of crickets using a weighing scale and place crickets in the respiration chamber / boiling tube. Place tubes in the water bath.

5. [1 mark Procedure] Allow 2 minutes for the cricket to adjust to the required temperature (equilibration).

6. Connect the manometer to the two boiling tubes as shown in the diagram.

7. Note the position of the manometer fluid and start the stopwatch.

8. After 10 minutes, read off the position of the manometer fluid against the scale.

9. Repeat steps 1 - 8 for 4 other temperatures, 35°C, 40°C, 45°C, 50°C, using cricket of similar mass and stage of growth (or same cricket(s)).

[1 mark Reliability and Reproducibility]

10. To ensure reproducibility of results, repeat steps 1 – 7 two more times for each temperature (triplicates: chemicals/apparatus should be from existing setup)

+

11. Repeat entire experiment 2 more times using new crickets and chemicals.

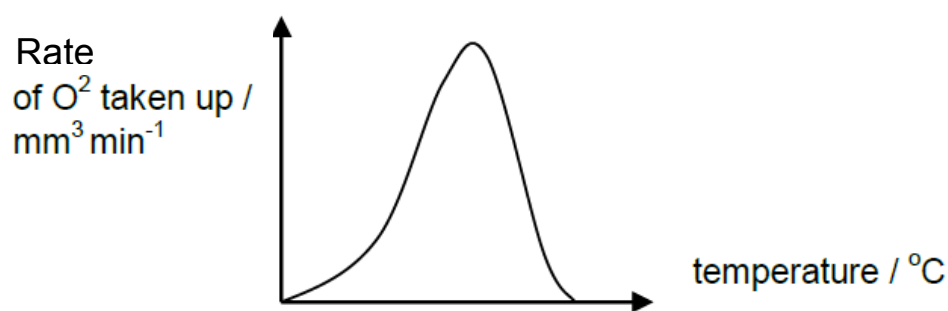
[1 mark Procedure]

12. Record the displacement of fluid in the manometer after 10 minute in a table. Since the carbon dioxide is absorbed by the soda lime, the change in volume is a result of oxygen uptake. Calculate the average rate of O₂ taken up in mm³ min⁻¹.

[1 mark for table]

Temperature/°C	Displacement of manometer fluid in 10 min / mm				Volume of O ₂ uptake / mm ³	Rate of O ₂ uptake / mm ³ min ⁻¹
	Replicate 1	Replicate 2	Replicate 3	Average		
30°C						
35°C						
40°C						
45°C						
50°C						

[1 mark for graph]

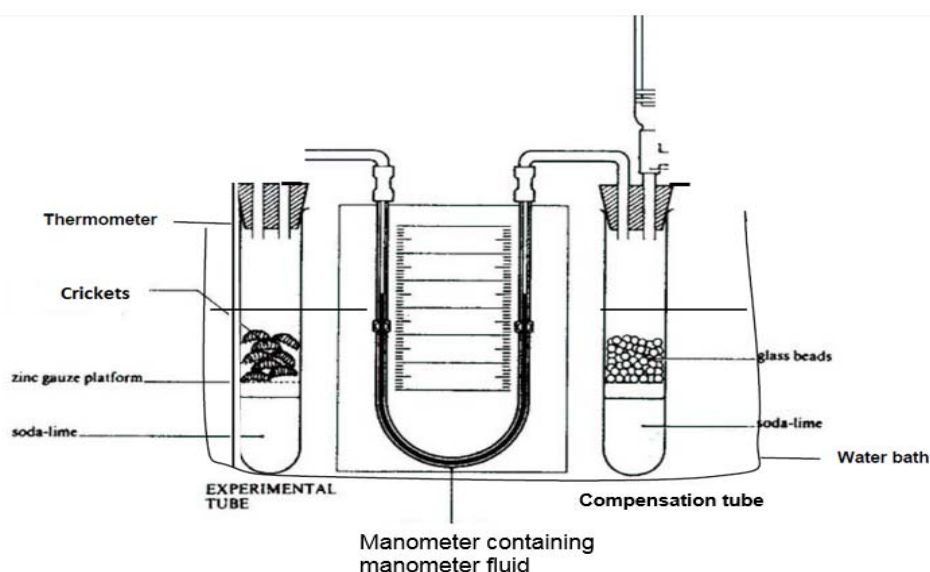
13. Plot a graph of rate of oxygen consumption (mm³ min⁻¹) against temperature (°C).

[1 mark for any 2 sets of Safety and Precautions]

14. Two safety precautions / risk hazards identified

- (a) Soda lime is toxic/corrosive, do not handle it with bare hands
- (b) Handle cricket with care to prevent bites/allergic reactions
- (c) Handle manometer with care to prevent breakages and cuts

[1 mark for labelled Diagram] Clamps to be included



Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labeled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5 **(a)** Discuss the advantages and limitations of the Polymerase Chain Reaction. [8]
- (b)** Explain how gel electrophoresis is used to analyse DNA. [5]
- (c)** Discuss the negative implications of Genetically Modified Organisms. [7]

[Total: 20]

5(a) Discuss the advantages and limitations of the Polymerase Chain Reaction.

.....[8]

ADVANTAGES OF PCR

- 1) Speed
 - **Millions of copies** of target DNA can be obtained in a relatively short amount of time
- 2) Ease of use
 - Easy to set up a PCR **reaction** (includes reagents in a tube) and the use of a thermocycler **machine**.
- 3) Sensitivity
 - PCR is capable of amplifying sequences from minute amounts of target DNA.
- 4) Specificity
 - Specificity in amplifying only target sequences as primers hydrogen-bond only to **complementary** sequences at opposite ends of the target segment
- 5) Robustness
 - A broad range of nucleic acid **sources** are suitable templates for PCR amplification.
- 6) Cell-free method of DNA replication
 - PCR is a cell-free method of DNA replication, and requires **no cleanup of unwanted cellular debris** or vector DNA.

[Any 4]**LIMITATIONS OF PCR**

- 1) Limited size range of DNA sequences cloned by PCR
Above this length, the polymerase tends to fall off.
/ The typical heating cycle does not leave enough time for polymerisation to complete.
- 2) High error rate
Due to lack of proofreading by Taq DNA polymerase / as Taq DNA polymerase lacks a 3' to 5' exonuclease activity.
- 3) Non-specific binding of primers (leads to non-specific amplification of sequences) due to sequence duplications (in the genome)
/ Partial primer binding could also occur, leaving the 5' end unattached.
- 4) Prior sequence information is usually necessary to construct the primers to bind to complementary sequences flanking the target sequence
- 5) Limited amounts of actual yields of PCR products compared to the amount that can be obtained using cell-based cloning where scale-up of the volumes of cell cultures is possible

[Any 4]

5(b) Explain how gel electrophoresis is used to analyse DNA.

.....[5]

- 1 DNA is negatively-charged; due to negatively-charged **sugar-phosphate backbone**;
- 2 move towards the **positively-charged electrode**;
through an agarose **matrix** which acts as a molecular sieve;
- 3 DNA fragments separated by size where shorter fragments move faster (reject: further) than longer ones;
- 4 Gel is stained with ethidium bromide ;
DNA fluoresce orange red under UV light;
- 5 Ethidium bromide intercalates between the bases of DNA;;
- 6 Size of the separated strands is **estimated** (Reject: determined) by comparison to the DNA ladder

5(c) Discuss the negative implications of Genetically Modified Organisms.

.....[7]
[1 mark for each well-elaborated idea: 2 bullet points]

1) Ownership issues [Social]

- Ownership issues – unfair for large multinational companies to **patent** GM plants.
+
- Increased dependence of undeveloped countries on rich developed countries. GM technology may benefit profit-driven research companies more than consumers in underdeveloped nations.

Animal rights issue [Ethical]

- Animal rights – GM animals may suffer unnecessarily;
+ 1 e.g.:
- Eg. The oncomouse has oncogene introduced into itself. The mice develop tumours more frequently than normal.
/ Increased use of the growth hormone may have harmful effects on animal health: additional stress of increased production of milk, meat and other products.

Possible application of gene cloning technique to humans [Ethical]

- One of the ethical concerns is that the cloning could be applied to humans; resulting in a eugenic generation of designer babies.

Meddling with biological processes [Ethical]

- Some groups or individuals see the use of GMO as intolerable **meddling with biological states or processes** that have naturally **evolved over long periods of time** [Reject: “playing God”]

Religious issues on GM foods [Ethical]

- religious implications in food choice especially when **GM foods are unlabelled**;
+
- Eg. incorporation of pig genes into plants/other animals;

Bt toxin resistant pests [Environmental]

- Pests may eventually become resistant to the Bt toxin
+
- farmers are required to plant at least 20% of non-Bt crops alongside Bt crops in refugia to support insect populations that are not under selection pressure and so slow development of resistance.

Unintentional harm to other organisms [Environmental]

- GM crops that kill pests could also cause unintentional harm to other organisms
+
- Eg. Pollen from Bt corn (not the BT toxin itself) could harm larvae from the Monarch butterfly.

Superweeds [Environmental]

- Seeds from **GM crops** might be carried to other places and **establish themselves as weeds**
+
- **Cross-pollination** between the GM crops and their **wild relatives** may spread the resistance to weeds

Disruption of ecological balance [Environmental]

- Ecological balance is disrupted
/ Accidental release of transgenic organisms into the environment might upset the balance of the ecosystem
+ 1.e.g:
- Fast-growing salmon may outcompete the wild salmon population and affect the food chain.
/ Larger transgenic salmon may be preferably selected as mates over smaller wild types.
/ Danger that the active growth hormone gene is transferred to other fish.

Human health and safety [Any 1]

- Introduction of foreign gene may result in production of secondary metabolites which may be toxic to animals themselves and/or livestock/humans that consume them.
/ New proteins in GM plants may be potentially allergenic to humans that consume them.
/ Vectors used in GE contain genes for antibiotic resistance. When these transgenic crops are eaten, these genes may pass from the plant to the *E. coli* in the gut, making them resistant to antibiotics.